Лабораторная работа №4

Выделение контуров

Вариант 7: оператор Прюитта 3×3.
Показаны: исходное изображение, полутон, матрицы Gx и Gy, модуль градиента G и бинаризация.
Порог бинаризации подобран опытным путём: T = 60.

Формулы

Полутон: I(x, y) = 0.299·R + 0.587·G + 0.114·B
Градиент: G = sqrt(Gx^2 + Gy^2)

Оператор Прюитта 3×3

Gx

-1	0	1
-1	0	1
-1	0	1

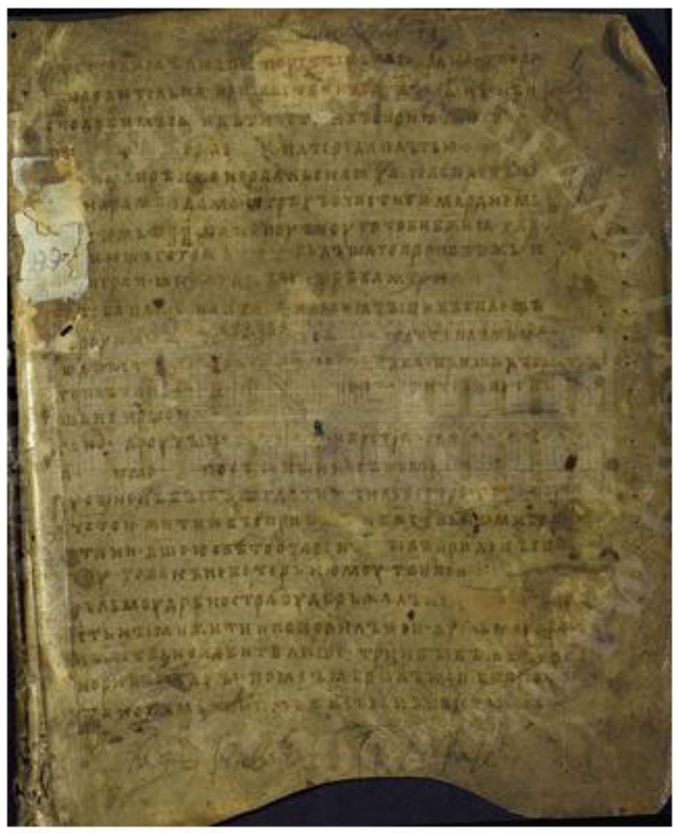
Gy

1	1	1
0	0	0
-1	-1	-1

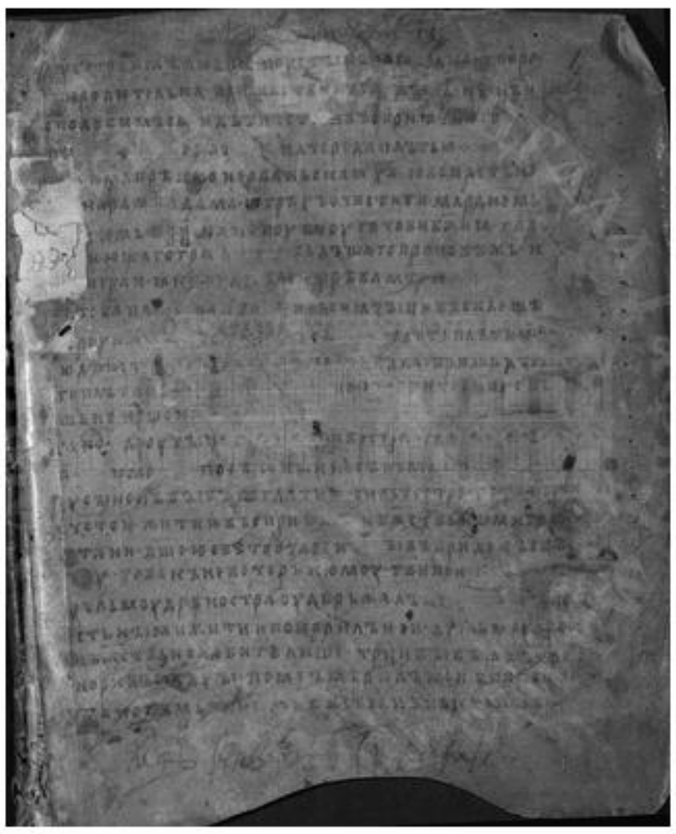
Матрицы Gx и Gy нормализованы по максимуму до диапазона 0…255.
Выходные файлы: *_gray.png, *_gx.png, *_gy.png, *_g.png, *_bin.png (папка output/).

Пример 1: slav_01.png

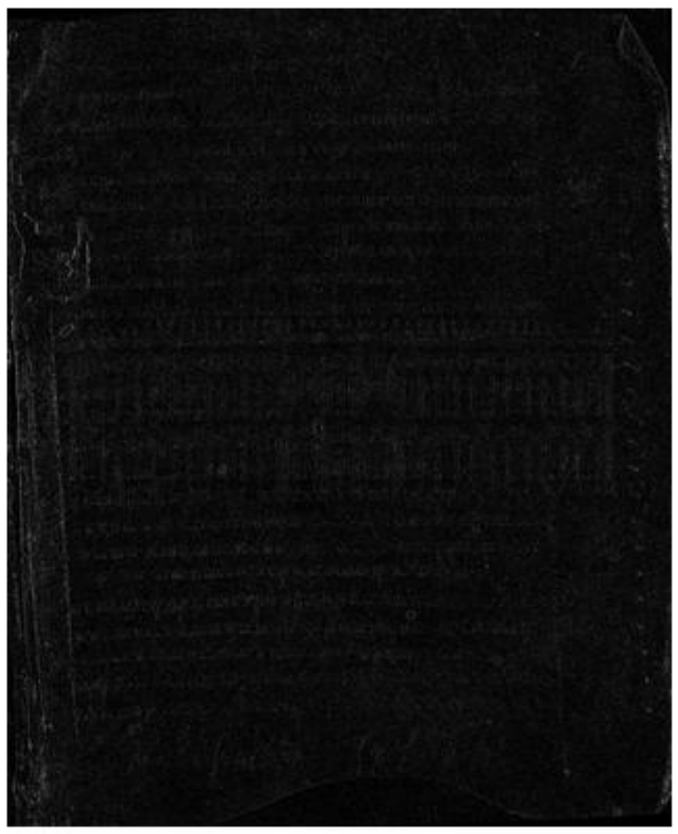
Исходное (RGB)



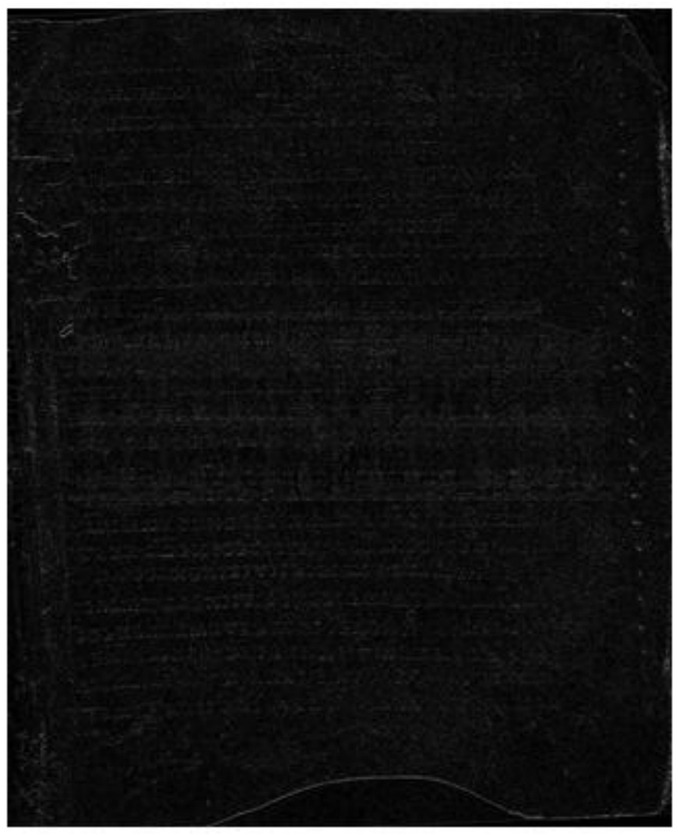
Полутон



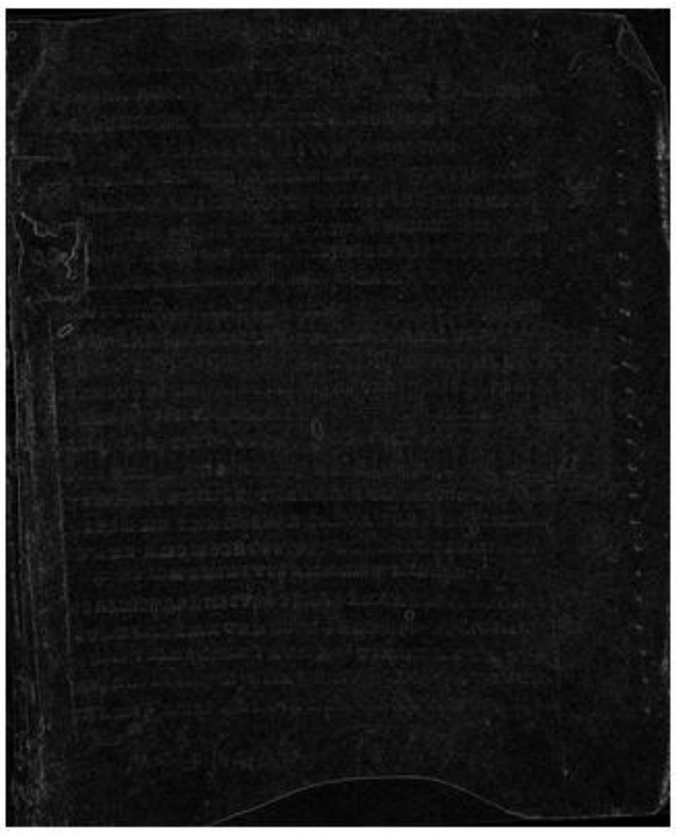
Gx



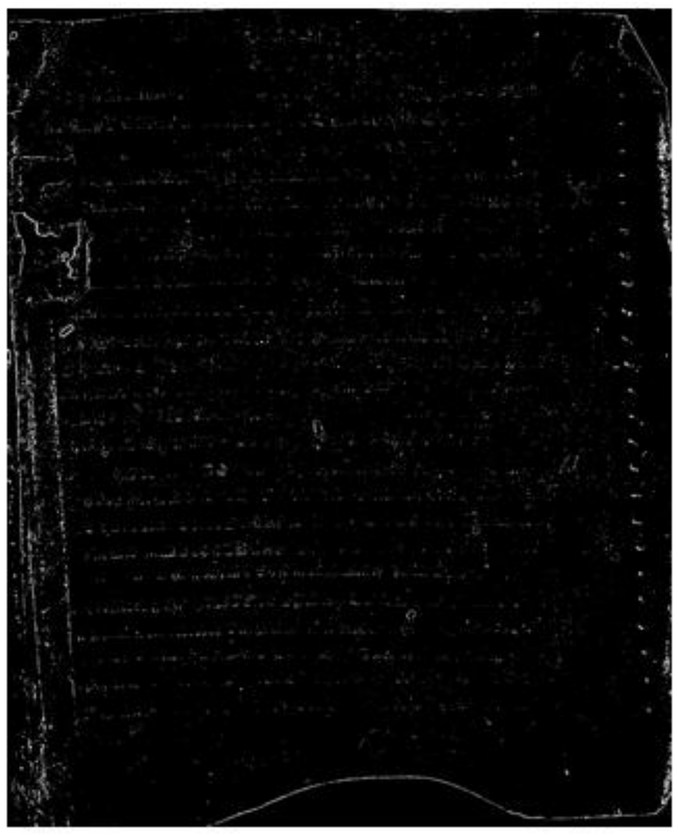
Gy



G

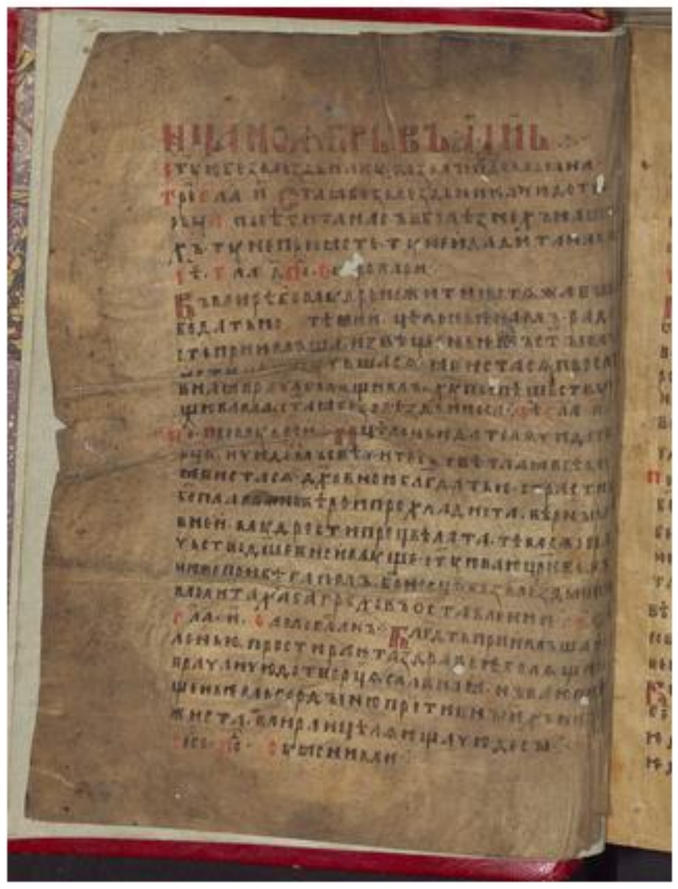


Бинаризация

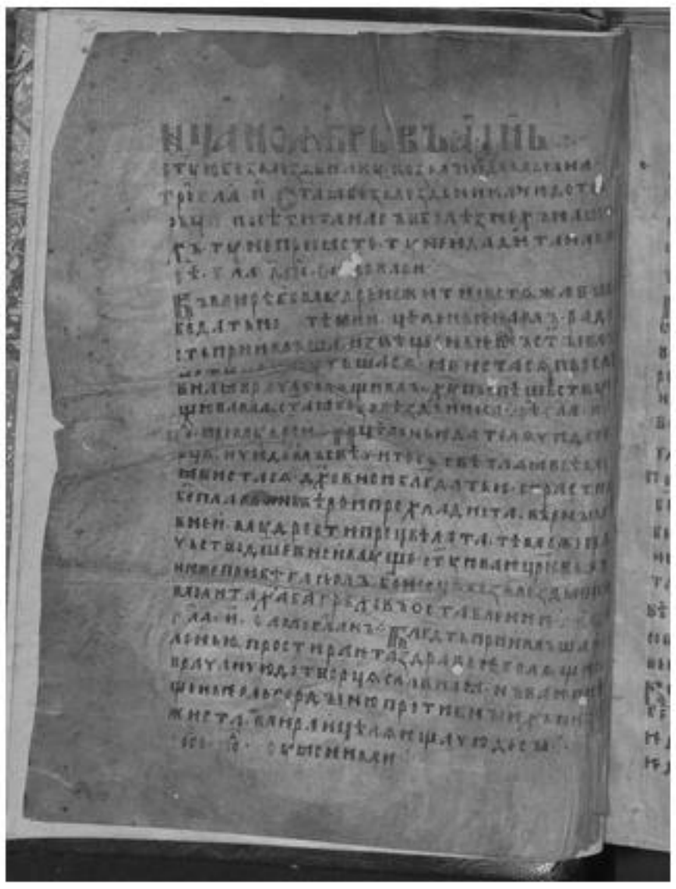


Пример 2: slav_02.png

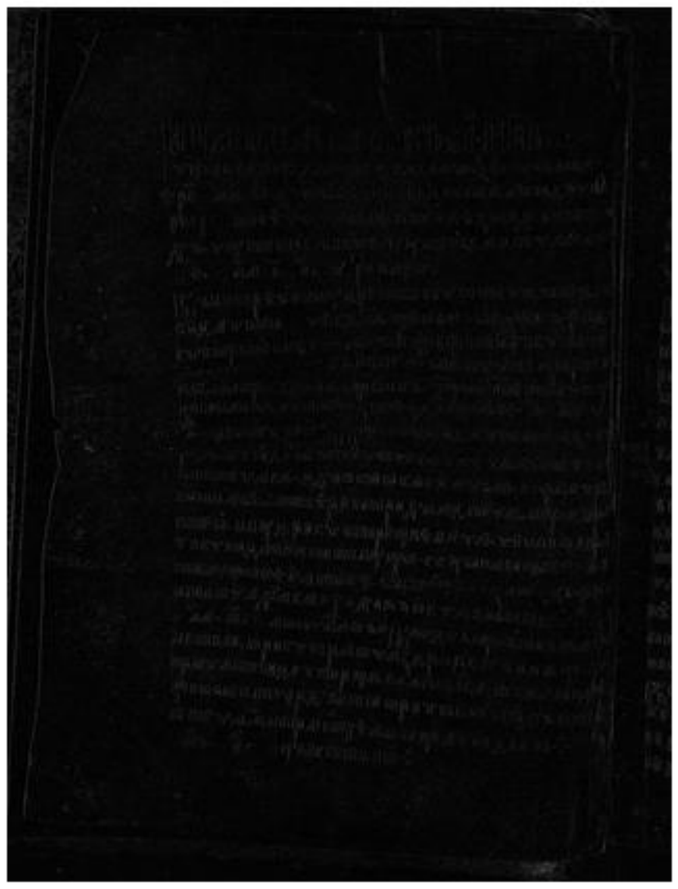
Исходное (RGB)



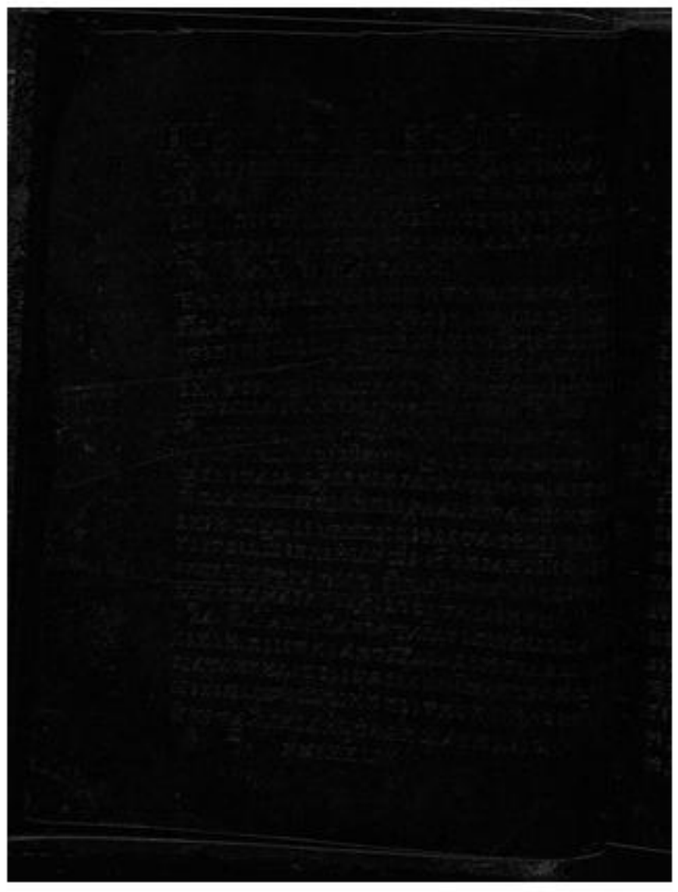
Полутон



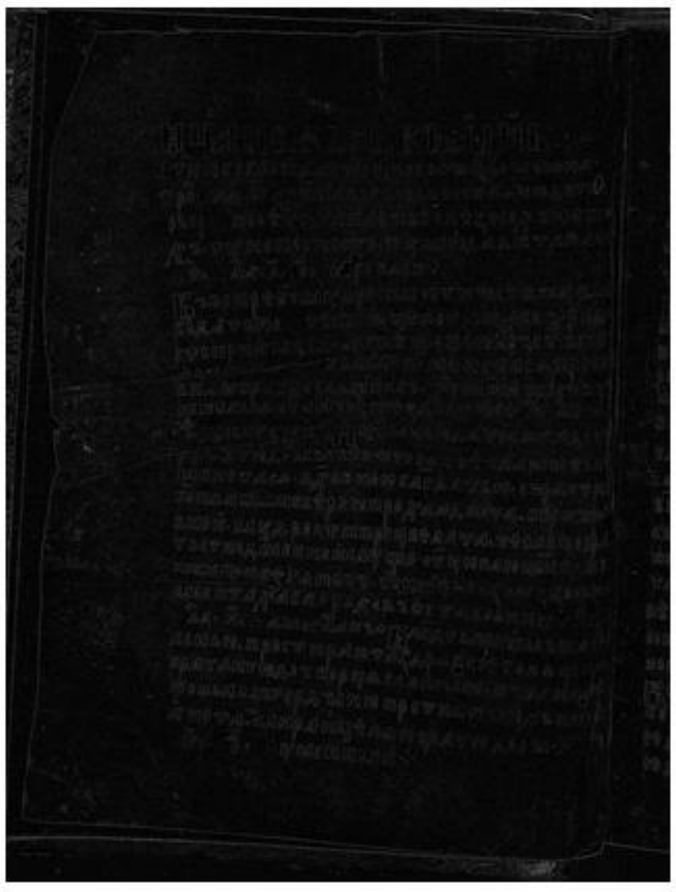
Gx



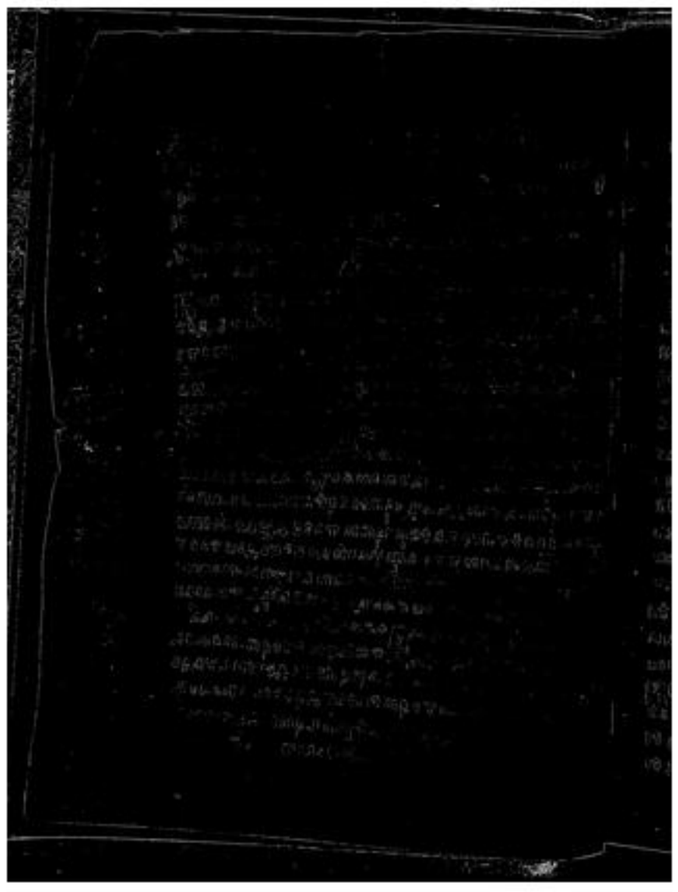
Gy



G



Бинаризация

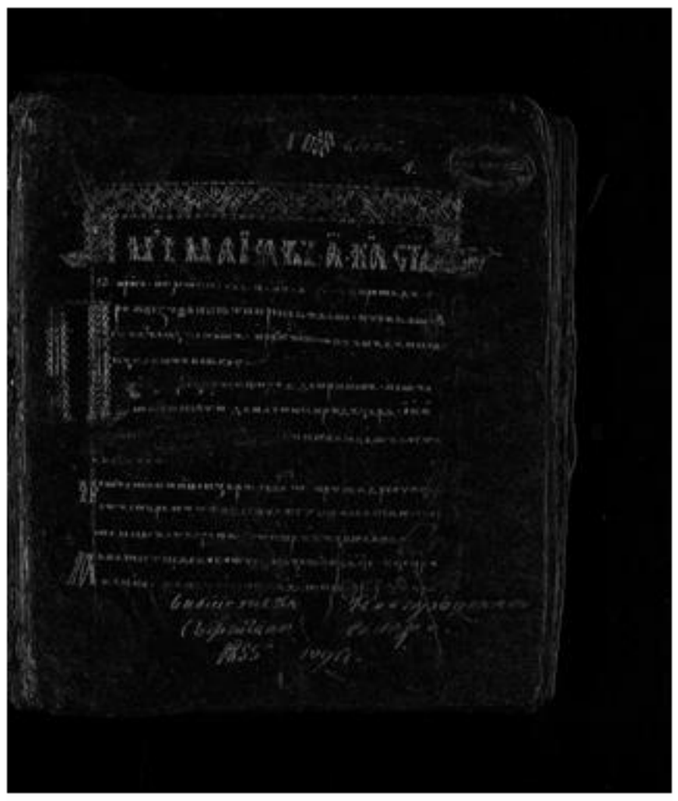


Пример 3: slav_03.png

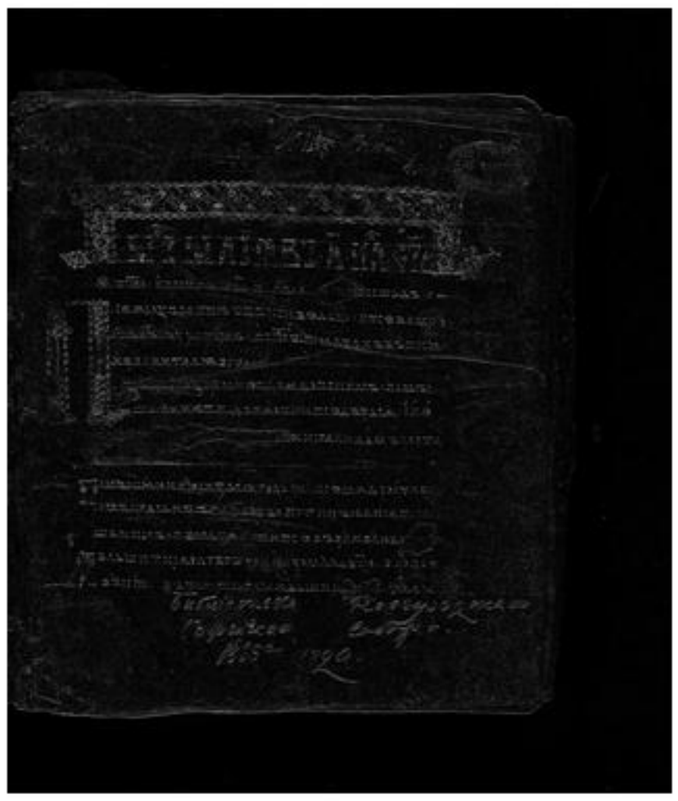
Исходное (RGB)

Полутон

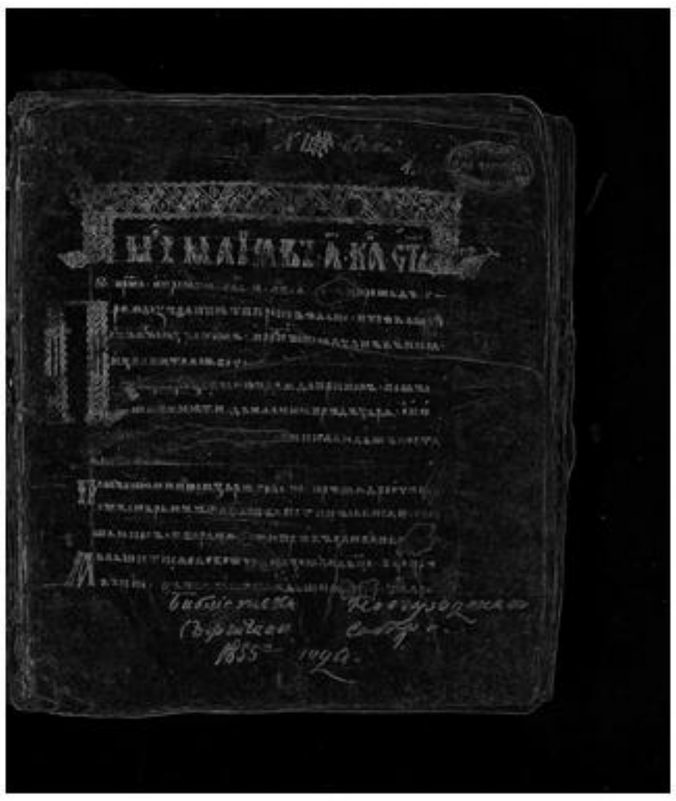
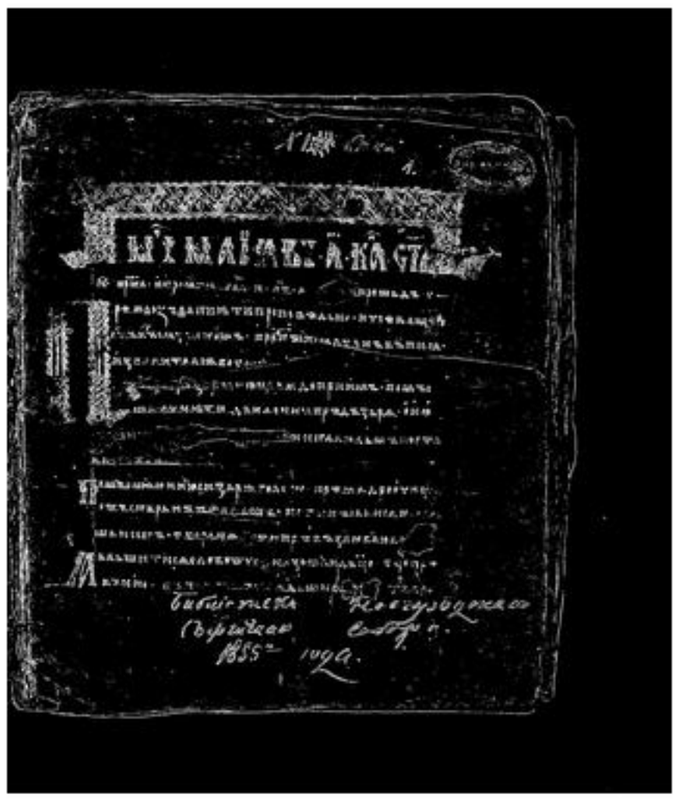
Gx



Gy



G



Бинаризация